

Ahmed Shaban

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Education

- 2018 – 2024 📖 **Doctor of Philosophy (PhD) in Physics**, North Carolina State University.
Thesis title: “*Spatially Resolved Galactic Outflows in Gravitationally Lensed Galaxies*”
Advisor: Prof. Rongmon Bordoloi (completed in Dec. 2024 and conferred on May 3, 2025).
- 2018 – 2020 📖 **Master of Science (MS) in Physics**, North Carolina State University.
- 2014 – 2018 📖 **Bachelor of Science (B.Sc.) in Physics**, Zewail City of Science and Technology, Egypt.
Concentration: *Astrophysics*.

Employment History

- 2025 – 📖 **Postdoctoral Fellow**, University of South Carolina.
Advisor: Prof. Varsha Kulkarni
- 2025 – 2025 📖 **Postdoctoral Research Scholar**, North Carolina State University.
Advisor: Prof. Rongmon Bordoloi
- 2019 – 2024 📖 **Graduate Research Assistant (RA)**, North Carolina State University.
- 2018 – 2022 📖 **Graduate Teaching Assistant (TA)**, North Carolina State University.
- TA & guest lecturer for the Solar system Astronomy class (110 Students). *Fall 2022*
 - Training students on using Telescopes in the Senior Physics Lab. *Fall 2021 & 2022*
 - TA for introductory physics curriculum development. *Summer 2022*
 - TA for the graduate Astrophysics class. *Spring 2022*
 - Instructor for E&M Lab (total: 152 Students). *Spring 2019 & Fall 2021*
 - Tutor at the Physics Tutorial Center (PTC). *Spring 2019*
 - Online tutor for E&M for Engineers and Scientists. *Fall 2018*
- 2018–2018 📖 **Teaching Assistant (TA)**, Zewail City of Science and Technology, Egypt.
- Teaching Assistant for PEU 331 (Stellar Structure & Evolution) *Spring 2018*

Invited Talks

- Feb. 2024 📖 “*Studying Galactic Winds via Gravitational Lensing*”, Invited talk at NC State University for visiting undergraduate students from UNC-Pembroke, Raleigh, NC.
- Oct. 2023 📖 “*Cosmic Lens on Galactic Winds*”, Invited talk at the Galaxies and AGN journal club at the Space Telescope Science Institute (STScI), Baltimore, Maryland.

Invited Talks (continued)

- *"Dissecting a 30 kpc Galactic Outflow"*, Invited talk at the Astro-coffee Journal Club at the Johns Hopkins University, Baltimore, Maryland.
- *"Spatially Resolving Galactic Outflows and the CGM using Gravitational Lensing"*, Invited talk at the Low-Density Universe subgroup at the Space Telescope Science Institute (STScI), Baltimore, Maryland.
- Apr. 2023 ■ *"Spatially Resolving Galactic Outflow at High- z "*, Invited Talk at Dr. Fabian Heitsch's group retreat at UNC-Chapel Hill, Durham, NC.

Observing

Observing Proposals

- (PI) **Gemini-North/GNIRS**, Semester 2026B 2026
 - Title: *"Unveiling the Nature of Damped Ly α Host Galaxies: NIR Spectroscopy of Nebular Emission at $z \sim 2$ "*
 - Program ID: GN-2026B-Q-222
 - Time Awarded: **20.4 hours** (band 2).
- (Co-Inv) **NASA-Keck Time/KCWI**, Semester 2024A 2024
 - Title: *"Spatially Resolved CGM metallicity maps at $z > 2$ "*
 - Program ID: 25/2024A_N110
 - Time Awarded: **2 nights**
 - PI: Rongmon Bordoloi

Observing Experience

- W. M. Keck Observatory, Keck II Telescope (Total: 11 nights):
 - Keck Cosmic Web Imager (KCWI): 10 nights.
 - Echellette Spectrograph & Imager (ESI): 1 night.

Mentoring

- 2025 – . . . ■ Laila Ahmed, Undergraduate Thesis project, University of Science and Technology at Zewail City of Science and Technology, Egypt.
- 2021–2024 ■ Ayesha Darekar, Undergraduate student research project, North Carolina State University, Co-advised with Prof. Rongmon Bordoloi.

Academic Service

- 2025 – . . . ■ Referee for the journal *Monthly Notices of the Royal Astronomical Society* "MNRAS".

Public Outreach

- 2019 – 2024  Co-organizer of the weekly Astrophysics journal club, NC State University.
- Oct. 2024  Co-organizer of star gazing event at NC State University.
- Feb. 2024  Juror at The 2024 US Invitational Young Physicists Tournament, Raleigh, NC.
- Jan. 2023  Volunteer at the Astronomy Days event at NC Museum of Natural Sciences.
- Sept. 2022  Organizing a stargazing event in Oak Island with Egyptian students from NC State University.
- Nov. 2019  Organizing an event to observe the 2019 Transit of Mercury at NC State University.

Skills

- Astrophysics Softwares  DS9, QFitsView, and Astropy.
- Operating Systems  Linux, Mac OS, and Windows.
- Web Dev  HTML and CSS.
- Programming  Python (Astropy, matplotlib, numpy, scipy, Pandas, Scikit-Learn, Tensorflow, Keras), Mathematica, Matlab, R, Java, Julia, and SQL.

Awards and Achievements

- Jun. 2022  **Graduate School Summer Fellowship**, North Carolina State University, \$2500.
- 2014-2018  **Merit-Based Scholarship** for my undergraduate studies at the University of Science and Technology at Zewail City of Science and Technology, Egypt.

Research Publications

Journal Articles

- 1 **A. Shaban**, R. Bordoloi, J. M. O'Meara, *et al.*, "Spatially Resolved Circumgalactic Medium Around a Star-Forming Galaxy Driving a Galactic Outflow at $z \approx 0.8$," *The Astrophysical Journal*, vol. 986, no. 2, 190, p. 17, Jun. 2025.  DOI: 10.3847/1538-4357/add0b9. arXiv: 2501.17940 [astro-ph.GA].
- 2 N. Giertych, **A. Shaban**, P. Haravu, and J. P Williams, "A statistical primer on classical period-finding techniques in astronomy," *Reports on Progress in Physics*, vol. 87, no. 7, 078401, p. 078 401, Jul. 2024.  DOI: 10.1088/1361-6633/ad4586. arXiv: 2205.10417 [astro-ph.EP].
- 3 **A. Shaban**, R. Bordoloi, J. Chisholm, *et al.*, "Dissecting a 30 kpc galactic outflow at $z \sim 1.7$," *Monthly Notices of the Royal Astronomical Society*, vol. 526, no. 4, pp. 6297–6320, Dec. 2023.  DOI: 10.1093/mnras/stad3004. arXiv: 2306.07328 [astro-ph.GA].
- 4 R. Bordoloi, J. M. O'Meara, K. Sharon, *et al.*, "Resolving the H I in damped Lyman α systems that power star formation," *Nature*, vol. 606, no. 7912, pp. 59–63, May 2022.  DOI: 10.1038/s41586-022-04616-1. arXiv: 2205.08554 [astro-ph.GA].
- 5 **A. Shaban**, R. Bordoloi, J. Chisholm, *et al.*, "A 30 kpc Spatially Extended Clumpy and Asymmetric Galactic Outflow at $z 1.7$," *The Astrophysical Journal*, vol. 936, no. 1, 77, p. 77, Sep. 2022.  DOI: 10.3847/1538-4357/ac7c65. arXiv: 2109.13264 [astro-ph.GA].

Conference Proceedings

- 1 **A. Shaban**, “Spatially Resolved Galactic Outflows and Circumgalactic Gas in Gravitationally Lensed Galaxies,” in *American Astronomical Society Meeting Abstracts*, ser. American Astronomical Society Meeting Abstracts, vol. 246, Jun. 2025, 2025n4i314p04, p. 314.04D.
- 2 **A. Shaban**, “Spatially Resolved Galactic Outflow at $z \sim 2$ Using Gravitational Lensing,” in *Oases in the Cosmic Desert: Understanding the Structure of the Circumgalactic Medium*, Arizona State University, Feb. 2023.
- 3 **A. Shaban**, R. Bordoloi, and J. O’Meara, “Small Scale Variation of Circumgalactic Medium Using Gravitational Lensing Tomography,” in *American Astronomical Society Meeting Abstracts*, ser. American Astronomical Society Meeting Abstracts, vol. 241, Jan. 2023, 327.01, p. 327.01.
- 4 A. Darekar, **A. Shaban**, R. Bordoloi, and J. O’Meara, “Probing the circumgalactic medium using a quadruply lensed quasar system,” in *American Astronomical Society Meeting Abstracts*, ser. American Astronomical Society Meeting Abstracts, vol. 54, Jun. 2022, 141.08, p. 141.08.
- 5 **A. Shaban** and R. Bordoloi, “A Spatially Resolved Study of Galactic Outflows in a Gravitationally Lensed Galaxy,” in *American Astronomical Society Meeting Abstracts #236*, ser. American Astronomical Society Meeting Abstracts, vol. 236, Jun. 2020, 307.01, p. 307.01.

References

Available Upon Request